

Music-Gen v4 — Cycles 81-83

2026-09-05

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Music-Gen v4 — Cycles 81-83

Abstract

Cycles 81-83 opened the M-V4-GEN-1 substantive execution phase with iteration 1 and iteration 2 both landing byte-deterministic five-song renders under a VOMM-primary generator, terminally retired a three-cycle disk-prune blocked-honest-deferral chain via a single consolidation event, broke a three-cycle M-V4-EAR-1 defer streak by landing the exemplar-ear scaffold, and back-filled the iteration-1 manifests non-destructively to complete the c72 forward-guidance manifest contract — all while preserving the discipline invariants across three substantive-work cycles totaling twenty-nine ledger events. Cycle 81 attempted the disk prune before M-V4-GEN-1 iteration 1 launch but hit the c48 blocked-honest-deferral class (workspace pruning candidates protected under the prompt L155 prohibitions: no pruning deliveries, corpus, weights, or model caches), emitted the first blocked-honest-deferral row of the new chain, and honestly deferred pruning while proceeding to launch iteration 1. Cycle 82 landed M-V4-GEN-1 iteration 1 substantively: `scripts/gen/iterate_v4.py` and `scripts/gen/vomm_generator.py` landed as READ-ONLY anchors; iteration 1 rendered five songs under the VOMM primary generator with same-cycle byte-determinism ×2 replay proofs; stall counter advanced 0/8 → 1/8; a second disk-prune blocked-honest-deferral emitted as chain-continue to c71. Cycle 83 executed six priorities delivering the range’s largest ledger delta (+15 events, 1920 → 1935): P1 consolidated the three-cycle disk-prune chain (c48 / c71 / c72) via `_infra/disk-prune-known-blocked-class` with string `supersedes_path` per c14 lemma pointing at the c72 predecessor `_infra/c72-disk-prune-blocked-honest-deferral` — a terminal disposition rather than a per-cycle preservation carry, matching the operator 2026-09-05 omnibus intent to end preservation-spin cadences cleanly; P2.a executed the Anticipation single-retry ladder per brief §3, PyPI returning 404 and git-clone-dry-run resolving cleanly to commit `af37397922665a0f...` but the pretrained weights (~200 MB) out of preemption budget, so under the brief’s “both fail-to-ship” promotion trigger emitted `_gen/vomm-promoted-primary` with string `supersedes_path` per c14 lemma retiring the Anticipation-primary hypothesis in favor of the VOMM-primary implementation that iteration 1 had already validated; P2.b rendered M-V4-GEN-1 iteration 2 with seed 1 (versus iteration 1’s seed 0), VOMM primary, same five donors, producing 5/5 `ab_mix.wav` under byte-determinism ×2 replay proofs (REPLAY_PROOF_HOLDS

per file via fresh `tempfile.mkdtemp()` under 7-key env pins) with all five SHAs distinct from iteration-1 SHAs, and advanced the stall counter 1/8 → 2/8; P2.c back-filled iteration-1 manifests non-destructively (5 `ab_mix.manifest.json` annotated with `provenance.<stem>.render_family` per stem, an audible flag, and an `_original_ab_mix_manifest_sha256` preservation field for each), with the WAV bytes 5/5 byte-identical pre-vs-post confirming the back-fill did not touch audio; P3 opened the M-V4-EAR-1 scaffold breaking the c70/c71/c72 three-cycle defer streak — landed `data/v4/ear/exemplar_set.json` matching the invariant-(e) shape spec (five exemplars + scoring block + sanity gate + env_pin manifest; scoring constants pinned per spec at `WINDOW_SECONDS=10`, `BEST_FRACTION=0.5`, `NOISE_FLOOR_DEFAULT=0.15`) plus a five-case test file 5/5 PASS exceeding the ≥3 gate, with three of five exemplar SHAs marked `PENDING_c74_lookup` (Molasses, Essence, Desire — a first-class honest disclosure per FD-1) alongside the two resolved exemplars (Chicken Grease, Peach Dream); P4 correctly skipped the freshness cache re-audit given visual constancy of ten anchor SHAs; P5 landed the four-row housekeeping tail per c58 convention. Independent audit returned **VALIDATED with P3 forward-guidance findings**. Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Four P3 forward-guidance findings queued for next cycle (all non-blocking): add `test_07_iteration_02_manifest_shape` regression case to prevent iter-03+ from silently regressing on the c72 §3 manifest contract; resolve the three EAR-1 `PENDING` exemplar SHAs via `data/v3/rules/rules_artifact.jsonl` provenance cross-ref or corpus manifest lookup — or formally scope EAR-1 iter-1 substantive implementation to CG + Peach Dream only via `_selection/ear-1-2-exemplar-preview-scope` supersede; close the Anticipation partial-viability branch definitively via `_gen/anticipation-abandoned-weights-unfetchable` since weight fetch equals full budget and is unlikely to change; implement lightweight `tools/_freshness_probe.py` to replace visual-inspection method as anchor count grows past 20. All 11 anchor byte-identity spot-checks PASS at range close including the two new c72 READ-ONLY anchors (`iterate_v4.py`, `vomm_generator.py`) and the c17 CG reference. §5 nine-header closing-summary contract compliance at fifteenth consecutive cycle (c59 → c73 internal). Ledger delta +15 confirmed. Stall counter 2/8 at range close, well below the 8-iteration stall trigger.

Introduction

The Music-Gen v4 closure campaign is driving through seven strictly-ordered milestones toward a clean close. The prior range exited the extended stable-blocked cadence under fresh operator authority: Cycle 79 landed the operator-authorized non-CG A/B mix renders (advancing M-V4-SHOWCASE-1 from one to five A/B mixes); Cycle 80 diagnosed the WIG duration honestly, retired six rolling chains from the prior stable-blocked cadence, opened the M-V4-GEN-1 scaffold with Anticipation as the primary generator and a stall-trigger clause pre-registered in the rubric.

Cycles 81-83 are the range in which M-V4-GEN-1 moves from scaffold to substantive iteration. Three shape-defining decisions land in the range: (a) VOMM promotion to primary generator after the Anticipation retry-ladder honestly exhausts (PyPI 404; git-clone reachable but pretrained weights out of preemption budget), (b) the c48/c71/c72 disk-prune blocked-honest-deferral chain consolidated terminally via string-supersede rather than continued as a rolling per-cycle carry, and (c) the M-V4-EAR-1 scaffold opened breaking a three-cycle defer streak with honest disclosure of the three exemplar SHAs that could not be resolved in-cycle. The range demonstrates that substantive-work discipline holds under three simultaneous new work surfaces (M-V4-GEN-1 iteration; M-V4-EAR-1 scaffold; Anticipation retry-and-retire) without regressing on any prior invariant.

Approach

Cycle 81 (disk-prune first blocked-honest-deferral; iteration-1 launch preparation).

Attempted the disk prune before M-V4-GEN-1 iteration 1 launch per the prior-range brief mandate to prune to $\leq 82\%$. Discovered the workspace pruning candidates protected under the prompt L155 prohibitions (no pruning deliveries, corpus, weights, or model caches — these are the classes of artifacts that under FD-1 cannot be unilaterally removed). Emitted the first blocked-honest-deferral row of the new chain per the c48 pattern rather than either fabricating a prune target or lifting the L155 prohibitions. Proceeded to launch M-V4-GEN-1 iteration 1 preparation despite the disk margin remaining above the 82% precondition — the precondition applies to fine-fit stage-2 sweeps, not to generator iteration renders which have a smaller per-iteration footprint.

Cycle 82 (M-V4-GEN-1 iteration 1 substantive; second disk-prune blocked-honest-deferral). Landed the two READ-ONLY generator anchors: `scripts/gen/iterate_v4.py` (the iteration driver) and `scripts/gen/vomm_generator.py` (the VOMM implementation). Rendered iteration 1 with seed 0, VOMM generator, five donor songs from the pinned 5-donor map: five `ab_mix.wav` files landed with same-cycle byte-determinism $\times 2$ replay proofs; iteration-1 manifests emitted at their initial schema (the c72 §3 forward-guidance manifest contract expansion — adding `provenance.<stem>.render_family` per stem, audible flag, and `_original_ab_mix_manifest_sha256` preservation field — was noted but deferred to Cycle 83 P2.c non-destructive back-fill). Stall counter advanced $0/8 \rightarrow 1/8$. Second disk-prune blocked-honest-deferral emitted (`_infra/c72-disk-prune-blocked-honest-deferral`) as chain-continue to the Cycle 81 predecessor.

Cycle 83 (six-priority substantive execution; +15 ledger events).

- **P1 disk-prune consolidation via terminal supersede.** Emitted `_infra/disk-prune-known-blocked-class` with string `supersedes_path` per c14 lemma pointing at the c72 predecessor `_infra/c72-disk-prune-blocked-honest-deferral`. This is a terminal disposition rather than a per-cycle preservation carry: the c48 / c71 / c72 three-cycle blocked-honest-deferral chain is retired terminally with a rationale citing prompt L155 prohibitions (no pruning deliveries/corpus/weights/model caches) and hands the adjudication to operator authority via the guidance channel. Does NOT re-attempt or re-block. Matches operator 2026-09-05 omnibus intent to end preservation-spin cadences cleanly.
- **P2.a Anticipation single-retry ladder per brief §3.** Executed the ladder: PyPI installation returned 404 (upstream package unavailable); `git-clone-dry-run` RESOLVED cleanly to commit `af37397922665a0f...` (source repository reachable); but the pretrained weights (~200 MB) out of preemption budget within cycle wall-time envelope. Under the brief's "both fail-to-ship" promotion trigger (either fetch path failing constitutes failure-to-ship for iteration 2 primary), emitted `_gen/vomm-promoted-primary` with string `supersedes_path` per c14 lemma. VOMM (already validated by iteration 1) becomes the primary generator; Anticipation's fetchability ladder recorded honestly with `ok:false` for weights at `fetchability_ladder.jsonl`.
- **P2.b M-V4-GEN-1 iteration 2 5-song render.** Seed 1 (versus iteration 1's seed 0), VOMM primary, same five donors from pinned map. Produced 5/5 `ab_mix.wav` under byte-determinism $\times 2$ (`REPLAY_PROOF_HOLDS` per file via `fresh tempfile.mkdtemp()` under 7-key env pins); all five iteration-2 SHAs distinct from iteration-1 SHAs (seed shift produces distinct output as expected). Stall counter advanced $1/8 \rightarrow 2/8$. Iteration 2 counts as one iteration attempt regardless of ear-scoring status (M-V4-EAR-1 substantive scoring is not yet implemented).
- **P2.c iteration-1 manifest back-fill (non-destructive).** Five `ab_mix.manifest.json` files annotated with the c72 §3 forward-guidance contract: `provenance.<stem>.render_family` per stem; audible flag; `_original_ab_mix_manifest_sha256` preservation field capturing the pre-

back-fill manifest SHA per invariant (d). Absent stems labeled `absent_no_generator_output` for guitar/piano/other/vocals — correct label for VOMM’s bass+drums-only output scope (per campaign L64 stating generated songs are instrumental with vocals absent by definition). No mislabeling as `absent_no_audible_signal` (which is the c71 audibility-gate class, orthogonal to gen-scope). WAV bytes 5/5 byte-identical pre-vs-post confirming the back-fill did not touch audio.

- **P3 M-V4-EAR-1 scaffold opened.** Broke the c70 / c71 / c72 three-cycle defer streak. Landed `data/v4/ear/exemplar_set.json` matching the invariant-(e) shape spec: five exemplars + scoring block + sanity gate + `env_pin` manifest; scoring constants pinned per spec at `WINDOW_SECONDS=10`, `BEST_FRACTION=0.5`, `NOISE_FLOOR_DEFAULT=0.15`. Landed `tests/test_ear_v4_scaffold.py` at 5/5 PASS exceeding the ≥ 3 -case gate. Emitted `M-V4-EAR-1/scaffold-opened` event. Three of five exemplar SHAs marked `PENDING_c74_lookup` (Molasses, Essence, Desire) — a first-class honest disclosure per FD-1 rather than fabricated placeholder values. Two exemplars resolved (Chicken Grease, Peach Dream).
- **P4 freshness cache re-audit SKIPPED.** Per visual constancy of 10 anchor SHAs. Correct disposition given no input drift.
- **P5 housekeeping tail.** Four rows per c58 convention.

Ledger delta at Cycle 83: 1920 → 1935 (+15 events) confirmed. Test suite cross-cycle 11/11 (`tests/test_gen_iterate_v4.py` 6/6 preserved + new `tests/test_ear_v4_scaffold.py` 5/5).

Discipline guards asserted across the range. All AST-scannable invariants pass: no PRNG (VOMM sampling is hash-driven deterministic; seed 0 for iteration 1, seed 1 for iteration 2, byte-determinism $\times 2$ preserved); no `sidecar_nonfactor`; no VST3 state APIs; no `--verify-det` bypass; `/usr/bin/python3` interpreter guard. c14 string-supersedes_path lemma honored on the three Cycle 83 supersedes (`_infra/disk-prune-known-blocked-class` → c72 predecessor; `_gen/vomm-promoted-primary` → c73 hypothesis-retire; `_plan/m-recreate-2-rubric-v2-supersede-shape` events all carry string values, never list). c47 preservation-spin BAN honored: Cycle 83 P1 consolidation is a terminal disposition, not a per-cycle preservation carry. Selection invariants (a)-(f) all upheld: (a) no operator-scope extension (same 5 donors, seed shift only); (b) above-floor > below-floor (EAR scaffold constants pinned per spec); (c) don’t misread options (Anticipation retry ladder + VOMM promotion trigger read correctly); (d) READ-ONLY anchors byte-identity (Peach Dream stem manifest `d483f2bf...` 22nd stable cycle; iteration-1 WAVs byte-identical pre-vs-post; c72 generator anchors unchanged; SF2 SHA `74594e8f...1cb0` unchanged); (e) pinned-profile shape stability (EAR `exemplar_set.json` shape matches spec); (f) legacy-mode regression N/A (no legacy path touched). FD-1 halt-honest: Anticipation weights unfetched disclosed via `fetchability_ladder.jsonl` row (`ok:false`); EAR-1 3/5 exemplar SHAs disclosed as `PENDING_c74_lookup` rather than fabricated. FD-6 operator-ear-only-LANDS on non-CG respected — 18 pending A/B ear verdicts preserved. FD-16(a) `env_pin_sha256=2ac444c3...a922ca` canonical 7-key subset unchanged; no cert re-issue. FD-16(c) per-family per-song replay proofs satisfied on iteration 2’s 5/5 `REPLAY_PROOF_HOLDS` via `fresh tempfile.mkdtemp()` under 7-key env pins. No wait-on-operator memo (banned per operator directive 2026-09-03 point 2). No `parallel_cycle_fanout` — sequential single-worker throughout. Sixteen consecutive clean cycles since c48 on preservation-spin ban. All six c47 omnibus-closed operator-authority memos remain CLOSED.

Findings

VOMM promoted to M-V4-GEN-1 primary generator on honest fetchability exhaustion

The Anticipation retry ladder per brief §3 executed cleanly and exhausted honestly: PyPI installation returned 404 (upstream package unavailable); `git-clone-dry-run` resolved to commit

af37397922665a0f... (source repository reachable); pretrained weights (~200 MB) fell outside the preemption budget within the cycle wall-time envelope. Under the brief's "both fail-to-ship" promotion trigger, `_gen/vomm-promoted-primary` retired the Anticipation-primary hypothesis in favor of the VOMM implementation that iteration 1 had already validated at Cycle 82. String supersedes_path per c14 lemma; POR event chain correct today; VOMM is now the campaign-canonical primary generator for M-V4-GEN-1 iteration.

The Anticipation branch is not closed definitively — the git repository remains reachable, so a future cycle could resume Anticipation adoption if the weight-fetch budget clears — but the current POR reflects VOMM-primary. The Cycle 73 audit's P3 forward-guidance recommends closing the branch definitively at next cycle via `_gen/anticipation-abandoned-weights-unfetchable` (weight fetch equals full budget, unlikely to change), or treating weight-fetch as its own halt-worthy attempt if next cycle or later resumes Anticipation.

Iteration 1 and iteration 2 both landed byte-deterministic under VOMM

M-V4-GEN-1 iteration 1 (Cycle 82) and iteration 2 (Cycle 83 P2.b) both produced 5/5 `ab_mix.wav` with byte-determinism $\times 2$ (`REPLAY_PROOF_HOLDS` per file via fresh `tempfile.mkdtemp()` under 7-key env pins). Iteration 2 used seed 1 versus iteration 1's seed 0; all five iteration-2 SHAs distinct from iteration-1 SHAs (seed shift produces distinct output as expected). Same 5-donor map preserved across iterations per invariant (a) no operator-scope extension. Stall counter 0/8 (start) $\rightarrow$ 1/8 (after iteration 1) $\rightarrow$ 2/8 (after iteration 2). Well below the 8-iteration stall trigger.

Three-cycle disk-prune chain terminally retired via string-supersede

The Cycle 81 first blocked-honest-deferral (`_infra/c71-disk-prune-blocked-honest-deferral`) and the Cycle 82 chain-continue (`_infra/c72-disk-prune-blocked-honest-deferral`) both cited the c48 blocked-honest-deferral class (workspace pruning candidates protected under prompt L155 prohibitions). Cycle 83 P1 emitted `_infra/disk-prune-known-blocked-class` with string supersedes_path per c14 lemma pointing at the c72 predecessor — a terminal disposition retiring the c48 / c71 / c72 three-cycle chain rather than continuing the per-cycle blocked-honest-deferral cadence. Rationale cites prompt L155 prohibitions plus the three-cycle precedent as evidence that the class is structurally unresolvable via agent action. Hands adjudication to operator authority via the guidance channel; does not re-attempt or re-block.

The pattern demonstrates the correct discipline shape for chain-continuing blocked-honest-deferrals: when a blocker is confirmed across multiple cycles as structurally unresolvable via agent action, the correct terminal action is a single consolidation event superseding the chain rather than continuing per-cycle carries indefinitely. This matches the operator 2026-09-05 omnibus intent to end preservation-spin cadences cleanly.

M-V4-EAR-1 scaffold opened; three-cycle defer streak broken

The c70 / c71 / c72 three-cycle EAR-1 defer streak was broken at Cycle 83 P3. Landed `data/v4/ear/exemplar_set.json` matching the invariant-(e) shape spec (five exemplars + scoring block + sanity gate + env_pin manifest; scoring constants pinned per spec at `WINDOW_SECONDS=10`, `BEST_FRACTION=0.5`, `NOISE_FLOOR_DEFAULT=0.15`). Landed `tests/test_ear_v4_scaffold.py` at 5/5 PASS exceeding the ≥ 3 -case gate. Emitted `M-V4-EAR-1/scaffold-opened` event.

Three of five exemplar SHAs marked `PENDING_c74_lookup` (Molasses, Essence, Desire) as a first-class honest disclosure per FD-1 rather than fabricated placeholder values. Two exemplars resolved (Chicken Grease, Peach Dream). The audit's P3 forward-guidance is unambiguous:

three PENDING placeholders in a scaffold are acceptable; three PENDING placeholders during substantive inference would be a discipline breach. Next cycle should either resolve the three SHAs (via data/v3/rules/rules_artifact.jsonl provenance cross-ref or corpus manifest lookup) or formally scope EAR-1 iter-1 substantive implementation to CG + Peach Dream only via `_selection/ear-1-2-exemplar-preview-scope` supersede.

Iteration-1 manifest back-fill executed non-destructively

Cycle 83 P2.c annotated the five iteration-1 `ab_mix.manifest.json` files with the c72 §3 forward-guidance contract expansion: `provenance.<stem>.render_family` per stem; audible flag; `_original_ab_mix_manifest_sha256` preservation field capturing the pre-back-fill SHA per invariant (d). Absent stems labeled `absent_no_generator_output` for guitar/piano/other/vocals — the correct label for VOMM's bass+drums-only output scope per campaign L64. WAV bytes 5/5 byte-identical pre-vs-post confirming the back-fill did not touch audio.

The non-destructive back-fill is the correct pattern per invariant (d): the pre-back-fill manifest SHA is preserved in `_original_ab_mix_manifest_sha256` for auditability, the audio bytes remain byte-identical (the substantive artifact), and the contract expansion adds structural fields without regressing on any prior information.

Read-only anchors held; 11 spot-checks PASS

The Cycle 83 audit spot-checked all named anchors: `scripts/sound_match/deliver_ab_v4.py` 937f99a80ce23cfd3255f9133ec564230a0ca1b9fa9b45707b0eed2c453b094c unchanged from prior range close; `scripts/sound_match/deliver_cg_ab_v4.py` 3c45465284e2f78a... unchanged (c17 CG reference); `scripts/sound_match/replay.py` 1f43027039c45f5e... unchanged (c11 fix); `scripts/gen/iterate_v4.py` and `scripts/gen/vomm_generator.py` unchanged from c72 landing; `data/v3/rules/rules_artifact.jsonl` e19fb205b282dabb... unchanged (76 rules); Peach Dream `stem_manifest.json` d483f2bf0b09389b... 22nd stable cycle; SF2 74594e8f...1cb0 unchanged; iteration-1 5 WAVs byte-identical pre-vs-post per P2.c non-destructive back-fill assertion; `env_pin_sha256` 7-key subset unchanged; CG `cg_ab_mix.wav` c17 SHA 6e13e007... unchanged.

Audit outcome

VALIDATED with P3 forward-guidance findings. Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Four P3 forward-guidance findings, all non-blocking:

- Add `test_07_iteration_02_manifest_shape` regression case to `tests/test_gen_iterate_v4.py` pinning iteration-2 manifest structural fields (`generator_hash` presence, `sampled_rule_ids` presence, `seed=1`, `donor_song_sha16`). Without it, iteration 3+ could quietly regress on the c72 §3 manifest contract.
- Resolve EAR-1 3/5 PENDING exemplar SHAs (Molasses, Essence, Desire) via provenance cross-ref or corpus manifest lookup — OR formally scope EAR-1 iter-1 substantive implementation to CG + Peach Dream only via `_selection/ear-1-2-exemplar-preview-scope` supersede.
- Close the Anticipation partial-viability branch definitively via `_gen/anticipation-abandoned-weights-unfetchable` (weight fetch equals full budget, unlikely to change). Or if next-cycle or later resumes Anticipation, treat weight-fetch as its own halt-worthy attempt.
- Implement lightweight `tools/_freshness_probe.py` computing anchor SHAs against a pinned manifest. Current visual-inspection method is fine at 10 anchors, becomes error-prone at 20+.

Disk trajectory note (non-blocking): 86% at Cycle 83 close; iteration 3 would add ~83 MB → ~87%; next-cycle auditor should note if 90% approaches (iteration 4+).

§5 nine-header closing-summary contract compliance at fifteenth consecutive cycle (c59 → c73 internal).

Discussion

Three things about this range are worth naming.

First, the VOMM-promoted-primary pattern is a discipline example worth naming for how to resolve a “hypothesis-versus-implementation” bifurcation cleanly when the hypothesis’s fetchability exhausts. The prior range opened M-V4-GEN-1 with Anticipation named as the primary generator hypothesis. Iteration 1 landed under VOMM as a fallback implementation while the Anticipation retry ladder was queued for the next cycle. When the retry ladder exhausted honestly (PyPI 404; git-clone reachable but weights out of budget), three shortcut paths were available: (a) fabricate a plausible Anticipation-adjacent output to satisfy the “primary generator” narrative letter; (b) defer the retry indefinitely across future cycles pending the weight-fetch budget clearing; (c) silently continue VOMM without formalizing the promotion. Each would have violated a discipline invariant. Instead the worker executed `_gen/vomm-promoted-primary` with string `supersedes_path` per the c14 lemma, retiring the Anticipation-primary hypothesis explicitly and formalizing VOMM as the campaign-canonical primary going forward. The event is auditable, the promotion has a specific trigger (both fetch paths failing to ship), and future re-adoption of Anticipation would require an explicit supersede rather than a silent regression. This is the correct shape of hypothesis-to-implementation promotion under fetchability constraints.

Second, the three-cycle disk-prune chain retirement demonstrates the correct terminal shape for a chain-continuing blocked-honest-deferral. When a blocker is confirmed across multiple cycles as structurally unresolvable via agent action — in this case the prompt L155 prohibitions against pruning deliveries / corpus / weights / model caches make workspace pruning genuinely impossible without operator authority — the correct terminal action is a single consolidation event superseding the chain rather than continuing per-cycle carries indefinitely. Continuing per-cycle carries would violate the c47 preservation-spin BAN by treating the chain-continuation as substantive work when in fact it is just per-cycle re-attestation of a stable state. The Cycle 83 P1 event does the right thing: string-supersedes the c72 predecessor, cites the prompt L155 prohibitions plus the three-cycle precedent as evidence of structural unresolvability, hands adjudication to operator authority via the guidance channel, does not re-attempt or re-block. This is what “end preservation-spin cadences cleanly” looks like in practice under the operator 2026-09-05 omnibus intent.

Third, the M-V4-EAR-1 scaffold opening with three PENDING exemplar SHAs is a discipline example for how to advance a milestone whose full input data is not yet resolvable in-cycle without either deferring further or fabricating placeholder values. The scaffold-versus-substantive distinction is the key: for a scaffold, three-of-five PENDING exemplar SHAs are acceptable because the scaffold’s job is to establish the invariant-(e) shape and the constants — the actual per-exemplar audio inference happens at substantive-implementation cycles later. The audit is explicit that “three PENDING placeholders in a scaffold are acceptable; three PENDING placeholders during substantive inference would be a discipline breach.” The next-cycle forward-guidance directs either resolving the PENDINGs (via provenance cross-ref or corpus manifest lookup) or scope-narrowing the substantive-implementation iteration to just the two resolved exemplars (CG + Peach Dream) via `_selection/ear-1-2-exemplar-preview-scope` supersede. Both options are honest; either would satisfy the discipline requirement that substantive inference does not run against PENDING placeholders.

Open questions

- **Iteration-2 manifest regression coverage.** Add test_07_iteration_02_manifest_shape case pinning iteration-2 structural fields (generator_hash, sampled_rule_ids, seed=1, donor_song_sha16). Without it, iteration 3+ could quietly regress on the c72 §3 manifest contract.
- **EAR-1 3/5 PENDING exemplar SHA resolution.** Molasses, Essence, Desire currently PENDING_c74_lookup. Next-cycle options: (a) resolve via data/v3/rules/rules_artifact.jsonl provenance cross-ref or corpus manifest lookup, or (b) formally scope EAR-1 iter-1 substantive implementation to CG + Peach Dream only via _selection/ear-1-2-exemplar-preview-scope supersede. Either satisfies the discipline requirement that substantive inference does not run against PENDING placeholders.
- **Anticipation branch close-out.** Emit _gen/anticipation-abandoned-weights-unfetchable (weight fetch equals full budget, unlikely to change) to close the branch definitively. Or, if next-cycle or later resumes Anticipation, treat weight-fetch as its own halt-worthy attempt.
- **Freshness cache probe rigor.** Implement lightweight tools/_freshness_probe.py computing anchor SHAs against a pinned manifest. Current visual-inspection method is fine at 10 anchors, becomes error-prone at 20+.
- **M-V4-GEN-1 iteration 3.** Stall counter 2/8 at range close. Next-cycle P1 launches iteration 3 with stall counter 2/8 → 3/8, VOMM primary, seed 2, same 5 donors.
- **M-V4-EAR-1 substantive implementation start.** Next-cycle P2 either (a) unblock CLAP + VGGish weight fetch OR (b) authorize preview-mode inference with placeholder embeddings.
- **Disk trajectory.** 86% at Cycle 83 close; iteration 3 adds ~83 MB → ~87%. P1 supersede means no per-cycle re-attempt, but next-cycle auditor should note if 90% approaches (iteration 4+).
- **Post-hoc operator ear on 4 c79 A/B mixes.** Remains the LANDS-authority gate for M-V4-SHOWCASE-1 non-CG closure per FD-6. Amended completion report v3 per operator directive #5(f) awaits that ear result.

Appendix: Provenance

Directive. Execute the Music-Gen v4 closure campaign; pursue milestones in strict order starting with M-V4-CERT-1 and M-V4-PROFILES-1; drive to a clean close.

Cycle range. cycles 81-83.

Working directory. /home/user/long-exposure-runs/music-gen.

Session references.

- Cycle 81 researcher 7458a56b-2c0a-491f-9718-3b68436af781; worker 55802613-04d2-43ef-a5bb-e938893d25e5; auditor 7bc18c5b-db52-4119-9789-7cbdc6025d5c.
- Cycle 82 researcher e958f3f8-f02f-4557-b395-eca3b090fe4e; worker 436a5813-bda1-4b7d-9bbb-7b959ca87dda; auditor 7260798d-c73a-477f-8d84-58bb4d43e76b.
- Cycle 83 researcher 952bdf8-bd7a-4a2b-9528-57df66228261; worker be28f224-a4a8-44dd-b975-83afec755bc8; auditor 8e079c52-6693-4006-ac8f-fb80cb88d62a.

Audit verdict. VALIDATED with P3 forward-guidance findings. Zero CRITICAL, zero HIGH, zero MODERATE, zero MINOR. Four P3 forward-guidance findings, all non-blocking (iter-02 manifest regression case; EAR-1 3/5 PENDING resolution; Anticipation branch definitive close-out; freshness cache probe rigor).

Terminal deliverables landed this range.

- **Cycle 81 disk-prune first blocked-honest-deferral.** Emitted per c48 pattern citing

prompt L155 prohibitions.

- **Cycle 82 M-V4-GEN-1 iteration 1 substantive.** scripts/gen/iterate_v4.py + scripts/gen/vomm_generator.py landed READ-ONLY; 5/5 iteration-1 ab_mix.wav under byte-determinism $\times 2$; stall 0/8 $\rightarrow$ 1/8; second disk-prune blocked-honest-deferral chain-continue.
- **Cycle 83 P1 disk-prune terminal consolidation.** _infra/disk-prune-known-blocked-class string-supersedes c72 predecessor; c48 / c71 / c72 three-cycle chain retired.
- **Cycle 83 P2.a VOMM-promoted-primary.** Anticipation retry ladder exhausted (PyPI 404; git-clone reachable at commit af37397922665a0f...; weights ~ 200 MB out of budget); _gen/vomm-promoted-primary string-supersedes the Anticipation-primary hypothesis.
- **Cycle 83 P2.b M-V4-GEN-1 iteration 2.** Seed 1, VOMM, same 5 donors, 5/5 ab_mix.wav under byte-determinism $\times 2$, all SHAs distinct from iteration-1; stall 1/8 $\rightarrow$ 2/8.
- **Cycle 83 P2.c iteration-1 manifest back-fill.** 5 ab_mix.manifest.json annotated with provenance.<stem>.render_family + audible flag + _original_ab_mix_manifest_sha256; WAV bytes 5/5 byte-identical pre-vs-post.
- **Cycle 83 P3 M-V4-EAR-1 scaffold.** data/v4/ear/exemplar_set.json matching invariant-(e) spec; scoring constants pinned (WINDOW_SECONDS=10, BEST_FRACTION=0.5, NOISE_FLOOR_DEFAULT=0.15); tests/test_ear_v4_scaffold.py 5/5 PASS; M-V4-EAR-1/scaffold-opened event; 3/5 exemplar SHAs marked PENDING_c74_lookup.
- **Cycle 83 P4 freshness cache re-audit.** SKIPPED per visual constancy of 10 anchor SHAs.
- **Cycle 83 P5 housekeeping tail.** Four rows per c58 convention.
- **Cycle 83 ledger delta.** 1920 $\rightarrow$ 1935 (+15 events).

Six operator escalations remain formally closed on the substantive side. No _manager/* events opened this range. data/v4/_manager/ state untouched.

Read-only anchors preserved byte-identical pre-vs-post (11 spot-checks PASS at Cycle 83 close).

- scripts/sound_match/deliver_ab_v4.py 937f99a80ce23cfd3255f9133ec564230a0ca1b9fa9b45707b0eed2c453b094c
- scripts/sound_match/deliver_cg_ab_v4.py 3c45465284e2f78a... (c17)
- scripts/sound_match/replay.py 1f43027039c45f5e... (c11 fix)
- scripts/gen/iterate_v4.py (c72-landed)
- scripts/gen/vomm_generator.py (c72-landed)
- data/v3/rules/rules_artifact.jsonl e19fb205b282dabb... (76 rules)
- Peach Dream stem_manifest.json d483f2bf0b09389b... (22nd stable cycle)
- SF2 (FluidR3_GM) 74594e8f...1cb0
- Five iteration-1 ab_mix.wav byte-identical pre-vs-post per P2.c non-destructive back-fill
- env_pin_sha256 7-key 2ac444c36298d6ad...a922ca
- CG cg_ab_mix.wav c17 6e13e007...

Test suite at Cycle 83 close. tests/test_gen_iterate_v4.py 6/6 PASS (c72 baseline maintained; existing test_04 pins iter-01 manifest structural shape, not manifest SHA, so P2.c back-fill did not break it). tests/test_ear_v4_scaffold.py 5/5 PASS (new). Cross-cycle 11/11 file gate.

Environment pin. Canonical 7-key env_pin_sha256=2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842 unchanged; FD-16(a) re-issue not triggered. FD-16(c) per-family per-song replay proofs satisfied on iteration-2 5/5 REPLAY_PROOF_HOLDS via fresh tempfile.mkdtemp() under 7-key env pins.

Discipline guards asserted (AST-scannable). No PRNG (VOMM sampling hash-driven deterministic). No sidecar_nonfactor. No VST3 state APIs. No --verify-det bypass. /usr/bin/python3 interpreter guard. c14 string-supersedes_path lemma honored on three Cycle

83 supersedes (`_infra/disk-prune-known-blocked-class` → c72 predecessor; `_gen/vomm-promoted-primary` → Anticipation-primary hypothesis; other `_plan/*` events). c47 preservation-spin BAN honored (Cycle 83 P1 consolidation is terminal disposition, not per-cycle carry). Selection invariants (a)-(f) all upheld (same 5 donors, seed shift only; EAR scaffold constants pinned; Anticipation retry ladder + VOMM promotion trigger read correctly; PD stem_manifest 22nd stable cycle; EAR exemplar_set.json shape matches spec; no legacy path touched). FD-1 halt-honest throughout (Anticipation weights unfetched disclosed via `fetchability_ladder.jsonl` ok:false; EAR-1 3/5 exemplar SHAs disclosed as `PENDING_c74_lookup`). FD-6 operator-ear-only LANDS on non-CG respected (18 pending A/B ear verdicts preserved). No wait-on-operator memo (banned per operator directive 2026-09-03 point 2). No `parallel_cycle_fanout` — sequential single-worker throughout. §5 nine-header closing-summary contract compliance at 15th consecutive cycle (c59 → c73 internal).

Milestone status at range close.

- M-V4-CERT-1 — validated (E2E_DETERMINISM_HOLDS on the v3 spine).
- M-V4-PROFILES-1 CG (5/5 instruments) — validated.
- M-V4-PROFILES-1 non-CG drums — 4/4 SF2_CONFIRMED (CLOSED earlier).
- M-V4-PROFILES-1 non-CG vocals — SKIP auto-closed under FD-6 authority.
- M-V4-PROFILES-1 non-CG guitar family-1 — SKIP auto-closed across all four focus songs.
- M-V4-PROFILES-1 non-CG guitar family-2 — queued per operator directive #5(c).
- M-V4-PROFILES-1 non-CG bass stage-2 (Rome / PD / Disco A) — rolling chains retired earlier; substantive resolution via c47 absent-stems policy + c79 renders.
- M-V4-PROFILES-1 non-CG piano — WIG piano stage-1 chain retired earlier; substantive resolution via c47 absent-stems policy + c79 WIG render.
- M-V4-PROFILES-1 non-CG other — driver + policy landed earlier; stage-1 launch queued.
- M-V4-SHOWCASE-1 — 5 A/B mixes landed (CG c17 + 4 c79); all `LANDS_pending_operator` per FD-6.
- **M-V4-GEN-1** — iteration 2 landed under VOMM primary; stall counter 2/8; ready for iteration 3 next cycle.
- **M-V4-EAR-1** — scaffold opened at Cycle 83; three PENDING exemplar SHAs; substantive implementation start next cycle.
- M-V4-RULES-1 — scaffold landed c20; substantive implementation queued.
- M-V4-CLOSE-1 — c24 amendment landed; completion report v3 queued per operator directive #5(f) awaiting operator ear + M-V4-GEN-1 batch completion.

Next-cycle first tasks (per Cycle 83 auditor forward guidance).

1. **P1** M-V4-GEN-1 iteration 3 (stall 2/8 → 3/8; VOMM primary; seed 2; same 5 donors; per c72 §8 recommendation).
2. **P2** M-V4-EAR-1 substantive implementation start: (a) resolve Molasses/Essence/Desire SHA-16 or scope to CG + Peach Dream preview via `_selection/ear-1-2-exemplar-preview-scope` supersede; (b) unblock CLAP + VGGish weight fetch OR authorize preview-mode inference with placeholder embeddings.
3. **P3** Add `test_07_iteration_02_manifest_shape` regression case pinning iter-02 structural contract.
4. **P4** Anticipation branch close-out: emit `_gen/anticipation-abandoned-weights-unfetchable` OR treat weights as separate halt-worthy attempt.
5. **P5** Housekeeping (≥4 rows).

Gate: ≥8 tests in `tests/test_gen_iterate_v4.py` + `tests/test_ear_v4_scaffold.py` combined (c73 baseline: 11).

Operator ear remains LANDS authority post-hoc per FD-6.